AN IDEMPOTENT ENDOMORPHISM SPLITS ITS OBJECT INTO THE KERNEL AND THE KERNEL OF ITS COMPLEMENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the statement is of the form 'constructed as in the statement of Lemma ...', so that statement (source lines 583-609) is delivered as a context span with role=context to keep the card self-contained.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Kernel and cokernel of an idempotent and of its complement. Same chapter, Lemma lemma-idempotent-kernel-cokernel, delivered on this card as a context span. Supplies the four morphisms constructed out of f and g = id - f and the identities p i = id and q j = id that the proof uses.
- Direct sums described by four morphisms. Same chapter, Remark remark-direct-sum. Supplies the characterization of a direct sum by four morphisms satisfying the identities, which is how the splitting is recognised.

Source reference. Original remark 0H9N: sources/stacks-homology/pinned-original-source.tex, source lines 575–575. Navigation pointer invoked at proof line 589; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original remark 0103: sources/stacks-homology/pinned-original-source.tex, source lines 372–372. Navigation pointer invoked at proof line 663; the counted span contains the pointer only and the cited result is not reproduced

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inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be a preadditive category, x be an object of $\mathcal{A}$, and $f : x \to x$ be idempotent. Write $g = id_x - f$ and recall by Remark 0H9N that $f \circ g = g \circ f = 0$.*

1. *If f has kernel $i : \mathrm{Ker}(f) \to x$ and p is the unique morphism such that $g = i \circ p$, then $p \circ i = id_{\mathrm{Ker}(f)}$ and p is the cokernel of f.*
2. *If f has cokernel $p : x \to \mathrm{Coker}(f)$ and i is the unique morphism such that $g = i \circ p$, then $p \circ i = id_{\mathrm{Coker}(f)}$ and i is the kernel of f.*
3. *If g has kernel $j : \mathrm{Ker}(g) \to x$ and q is the unique morphism such that $f = j \circ q$, then $q \circ j = id_{\mathrm{Ker}(g)}$ and q is the cokernel of g.*
4. *If g has cokernel $q : x \to \mathrm{Coker}(g)$ and j is the unique morphism such that $f = j \circ q$, then $q \circ j = id_{\mathrm{Coker}(g)}$ and j is the kernel of g.*

Lemma 0.2. *Let $\mathcal{A}$ be a preadditive category, x be an object of $\mathcal{A}$, and $f : x \to x$ be idempotent. Write $g = id_x - f$. If each of f and g have at least one of a kernel or cokernel and in turn morphisms i, j, p, q are constructed as in the statement of Lemma 0.1, then all four (co)kernels exist and*

$$\begin{aligned} x &\simeq \mathrm{Ker}(f) \oplus \mathrm{Ker}(g) \\ &\simeq \mathrm{Coker}(f) \oplus \mathrm{Ker}(g) \\ &\simeq \mathrm{Ker}(f) \oplus \mathrm{Coker}(g) \\ &\simeq \mathrm{Coker}(f) \oplus \mathrm{Coker}(g) \end{aligned}$$

with the structure of each direct product given by i, j, p, q.

Proof. By Lemma 0.1 $p \circ i = \mathrm{id}_{\mathrm{Ker}(f)} = \mathrm{id}_{\mathrm{Coker}(f)}$ and $q \circ j = \mathrm{id}_{\mathrm{Ker}(g)} = \mathrm{id}_{\mathrm{Coker}(g)}$. By construction, $i \circ p + j \circ q = f + g = \mathrm{id}_x$. By Remark 0103 the claim follows. □